5.0 IN-LAB IMPEMENTATION & MEASUREMENTS (5 marks in total):

Part I: The Step Response of a Second-Order Bandpass Circuit

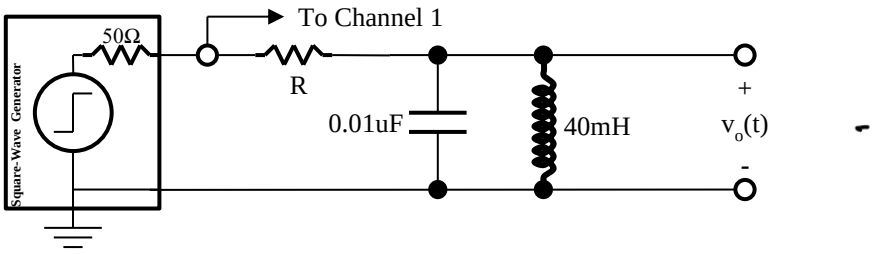


Figure 5.0: Second-Order Bandpass Filter

(a) Step 1: Connect Channel (1) of the oscilloscope to display the open-circuit voltage $v_s(t)$ of the function generator. Set the following:

- Trigger: source→Channel (1), and slope→rising.

Adjust the controls of the function generator to provide a square-wave signal $v_s(t)$ with a peak-to-peak value of 10V and DC offset of 5V at a frequency of 20Hz.
For ENG301, 302, 303 (FG): Waveform → Square, Frequency → 20 Hz, Amplitude → 5 Vpp*, Offset → 2.5 V*

*Due to FG matching impedance (EDU33212A)

For ENG402 (FG): Directly connected it with Oscilloscope, adjust AMPL knob to get 5 Volt peak (Vp) and adjust the OFFSET knob (you may need to pull it out), to get +5V DC off set. *Function Generator GFG8216A

Next, set the following:

- Channel (2): Vertical-position→one division above the bottom of the screen, coupling→dc, and V/div→1V.
- Time: Time/div→50μs when $v_s(t)$ is rising.
- Blank off Channel (1).

(b) Step 2: Construct the circuit shown in **Figure 5.0**. Set R=3.4kΩ. Connect Channel (2) to display the step response $v_o(t)$. Plot $v_o(t)$ on **Graph 1.0.**

(c) Step 3: Use the cursors on the oscilloscope to measure points on the $v_o(t)$ display to calculate the parameters (σ and ω) that characterize the step response as:

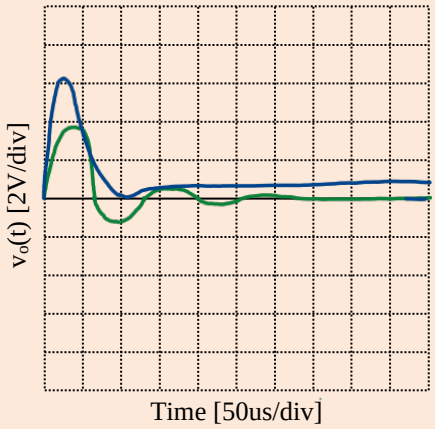
$$\mathbf{v_o}(t) = A\boldsymbol{e}^{-\boldsymbol{\sigma t}}\sin(\omega t)$$

(d) Step 4: Record your results in **Table 1.0.**

(e) Step 5: Demonstrate the correct operation of your setup to your TA. (1 mark)

(f) Step 6: Set R=1.4kΩ, and repeat as in Step 2 and Step 3. Record your results in **Table 1.0.**

(g) Step 7: Set R=500Ω, and repeat as in Step 2.



Graph 1.0 (0.5 marks)

Table 1.0 (0.5 marks)

R (KΩ)	T	$v_o(T/4)$	$v_o(3T/4)$	σ	ω
3.4	127 µs	3.63 V	-1.04 V	19 685.2	49473.9
1.4	136µs	5.51 V	300 mV	42 802	46 199.4

Part II: The Step Response of a Second-Order Lowpass Circuit

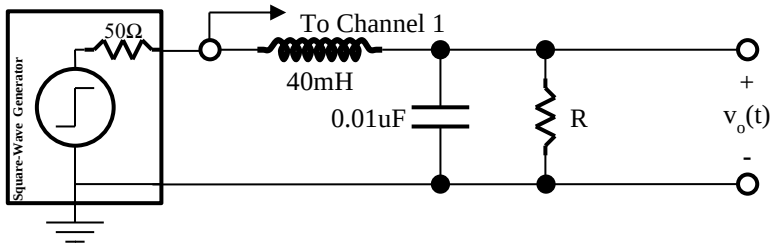


Figure 6.0: Second-Order Lowpass Filter

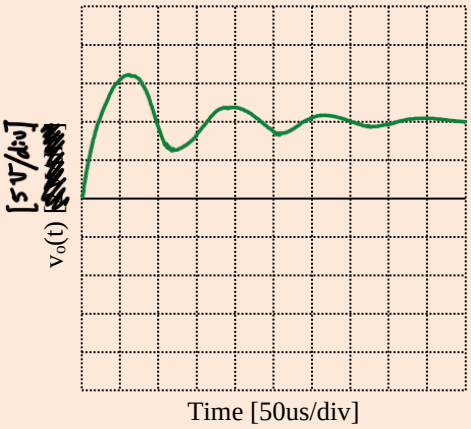
(h) Step 8: Connect the circuit shown in **Figure 6.0.** Set R=10kΩ. Use Channel (2) to display the step response v_o(t). Plot v_o(t) on **Graph 2.0.**

(i) Step 9: Use the cursors on the oscilloscope to measure points on the v_o(t) display to calculate (and record in **Table 2.0**) the parameters (σ and ω) that characterize the step response as:

$$\mathbf{v_o}(t) = B + Ae^{-\sigma t}\cos(\omega t + \theta)$$

(j) Step 10: Set R=1.4kΩ, and repeat as in Step 8.

(k) Step 11: Set R=1kΩ, and repeat as in Step 8.



Graph 2.0 (0.5 marks)

Table 2.0 (0.5 marks)

R (KΩ)	T	$v_o(T/4)$	$v_o(3T/4)$	σ	ω
10	127 μs	15.85	6.21	7179	99473.9

Part III: The Step Response of a Second-Order Highpass Circuit

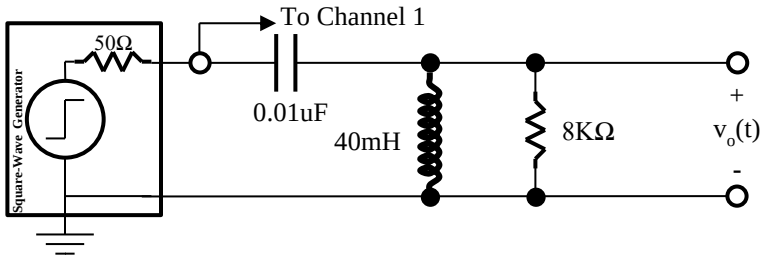


Figure 7.0: Second-Order Highpass Filter

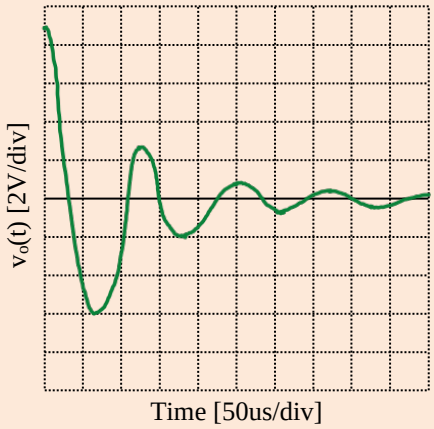
(l) Step 12: Connect the circuit shown in **Figure 7.0.** Connect Channel (2) to display the step response $v_o(t)$. Plot $v_o(t)$ on **Graph 3.0.**

(m) Step 13: Use the cursors on the oscilloscope to measure points on $v_o(t)$ display to calculate (and record in **Table 3.0**) the parameters (σ and ω) that characterize the step response as:

$$\mathbf{v_o}(t) = Ae^{-\sigma t}\cos(\omega t + \theta)$$

(n) Step 14: Demonstrate the correct operation of your setup to your TA. (1 mark)

0



Graph 3.0 (0.5 marks)

Table 3.0 (0.5 marks)

R (KΩ)	T	v_o(T/4)	v_o(3T/4)	σ	ω
8	128 μs	-6.04	3.06	10 624.8	49 087.4

6.0POST-LAB QUESTIONS (2 marks in total, 2/3 marks for each question):

(1) By examining your plots on **Graph 1.0,** answer the following:

a) What are the effects of varying the value of R on the step response of a second-order bandpass circuit?

A longer R gave a smaller σ value (19.7k vs. 42.8k) which meant the circuit is less damped. This allowed the oscillation to last longer as the decay was smaller.

A smaller R gave the opposite. A smaller σ value which meant the circuit is more damped. This lead to the graph to decay faster.

Furthermore, the frequency increases slightly as R increases and therefore the period is shorter.

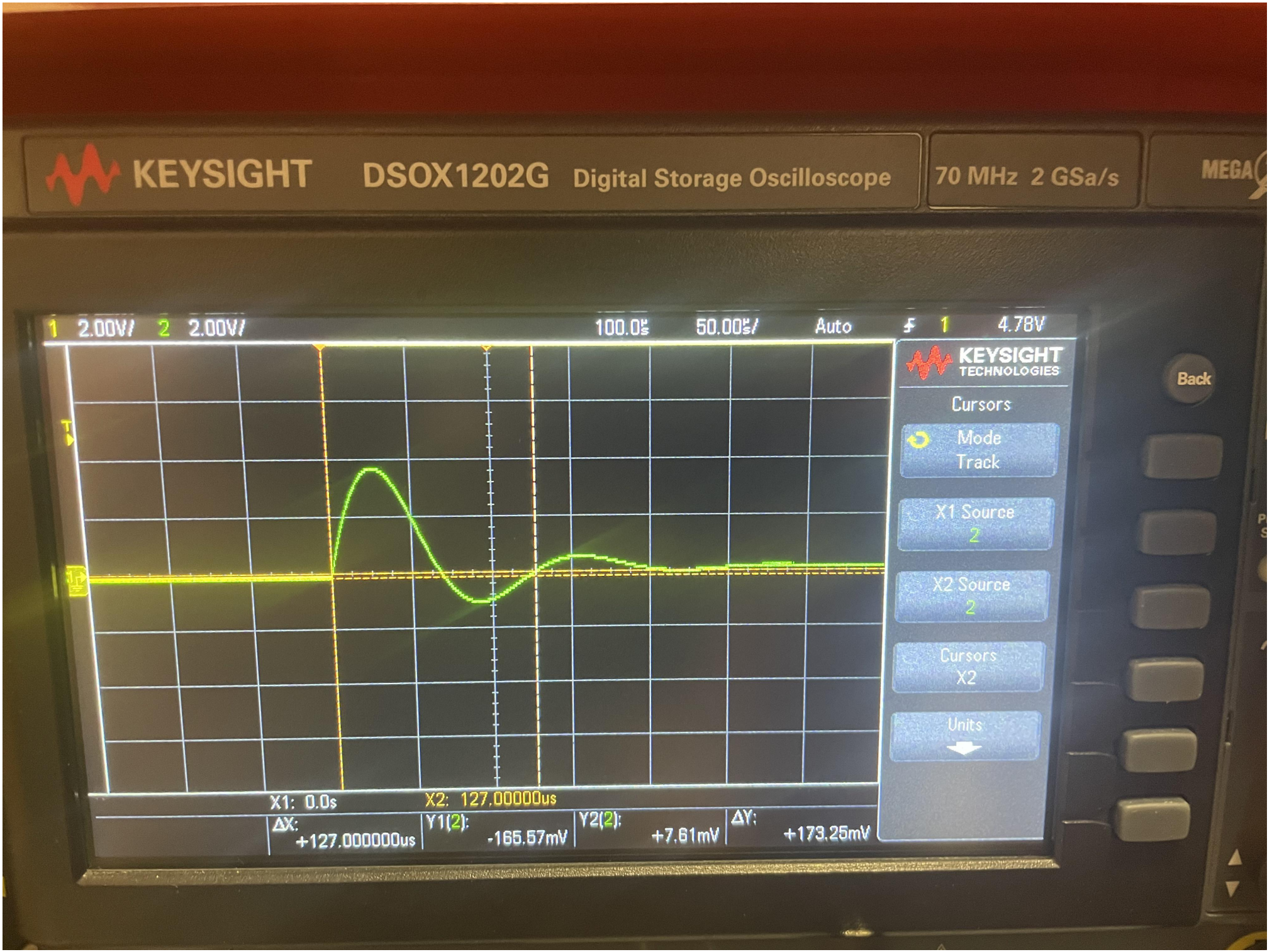
(2) By examining your plots on **Graph 2.0,** answer the following:

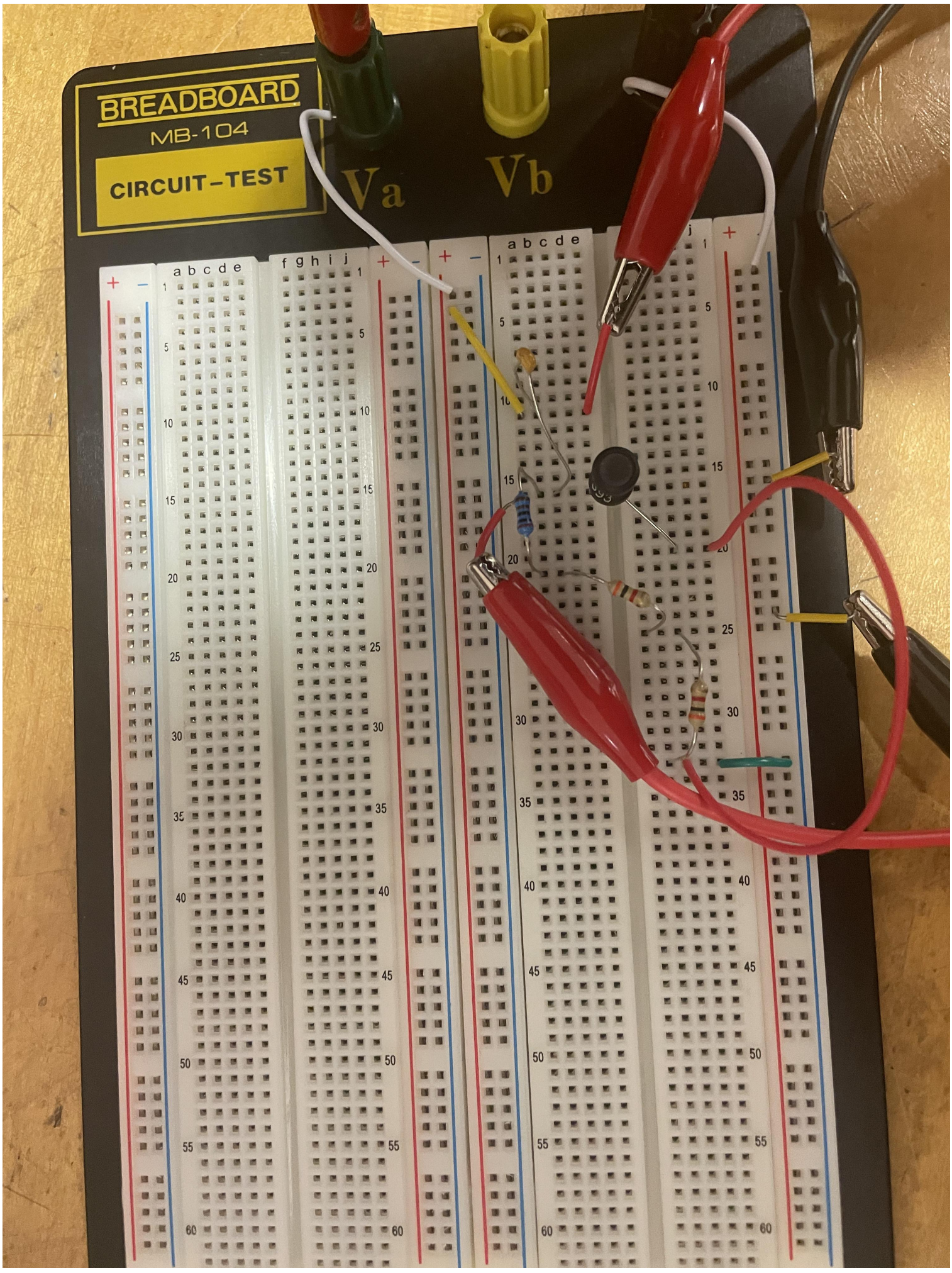
a) What are the effects of varying the value of R on the step response of a second-order lowpass circuit?

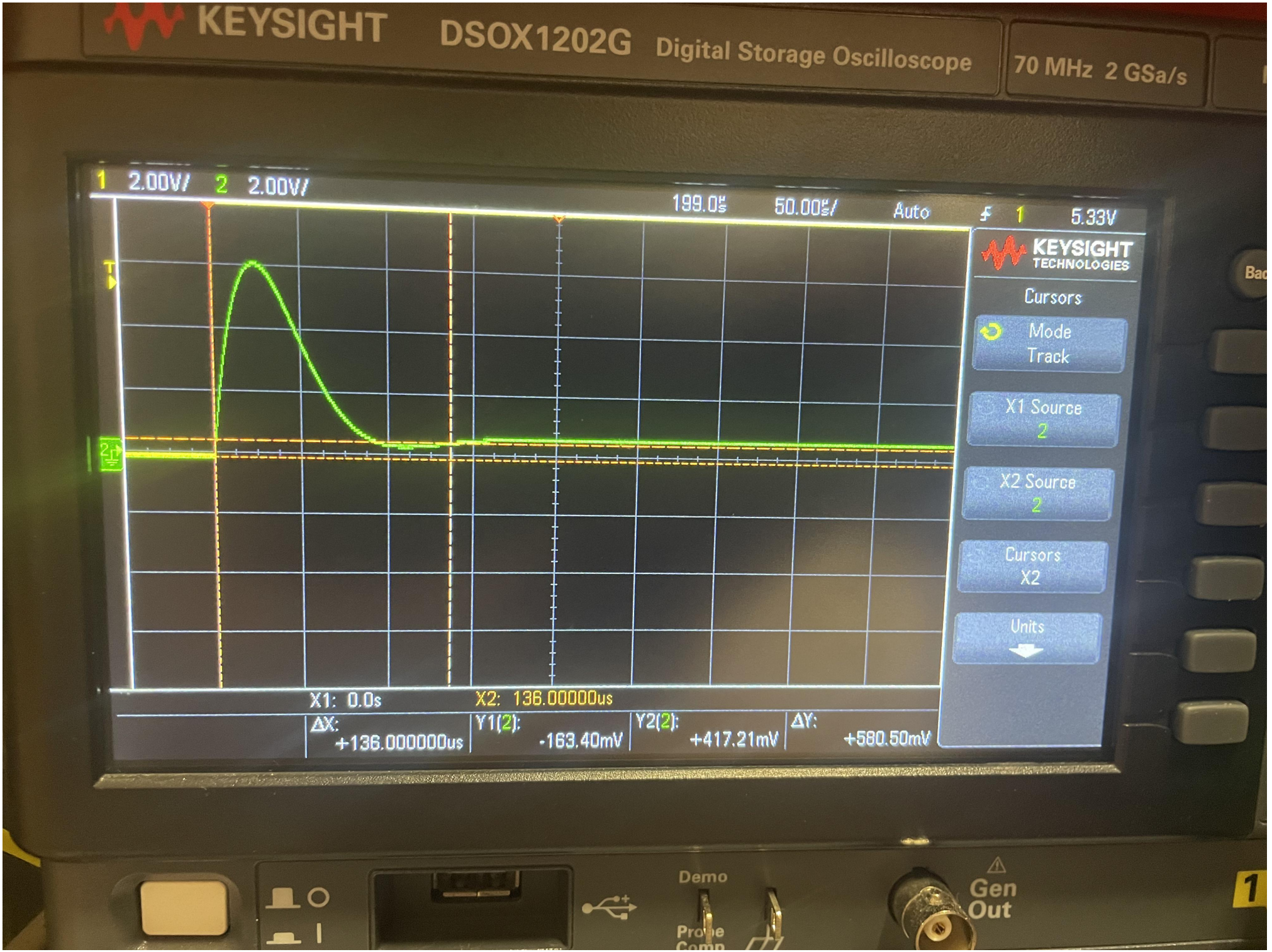
A larger R gives a smaller σ value. At 10k, the response was underdamped so it goes well past the final value (15.85V vs. 9.95V) and oscillates several times before settling. Decreasing R lowers σ which dampens the circuit more and causes the overshoot to shrink as well as shrinks the oscillation. Any resistance under around 1kΩ will be below critical dampening ($\sigma = \omega_0$) and the circuit will rise without overshoot.

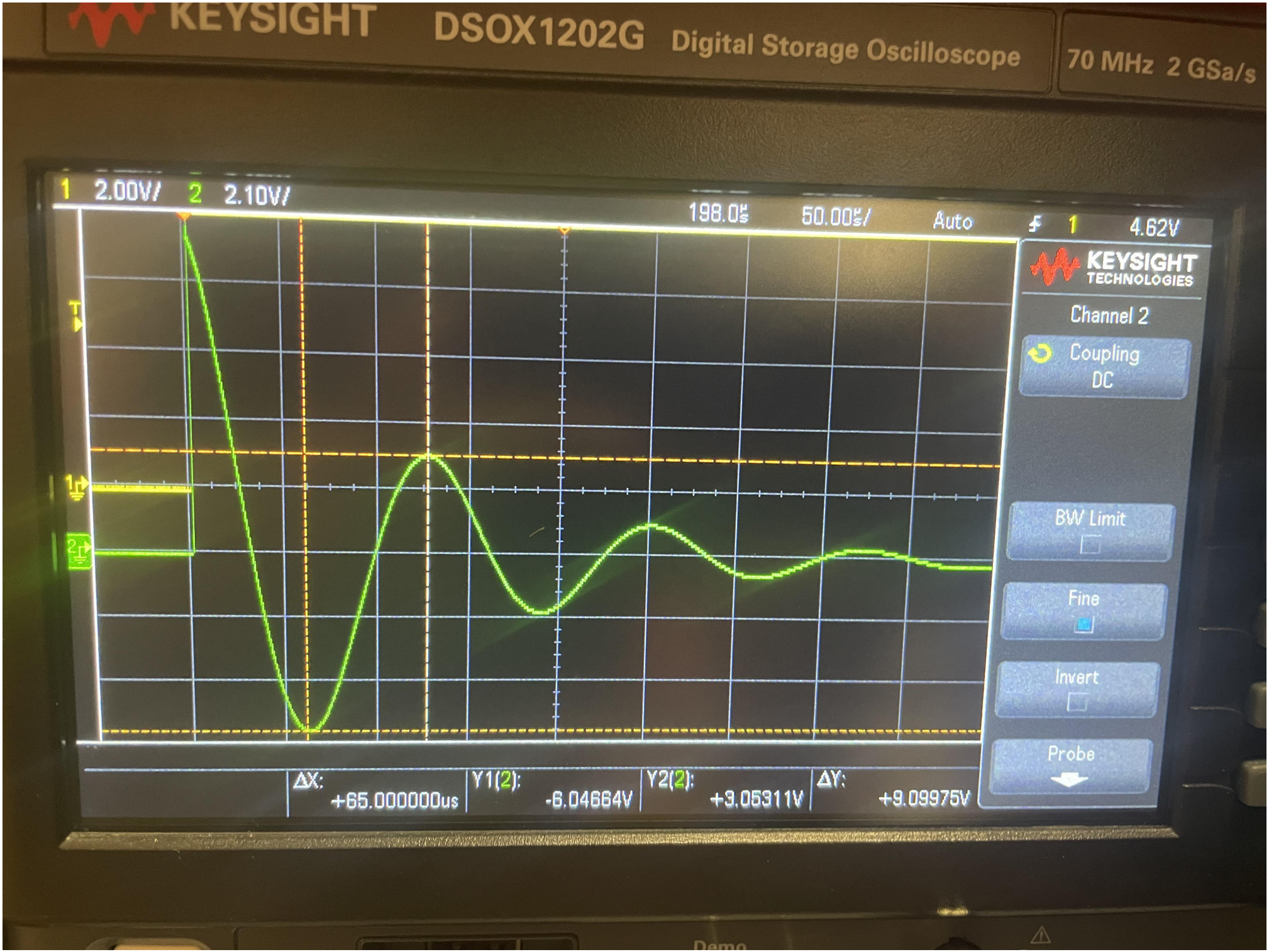
(3) Suppose that the 8kΩ-resistor is removed from the circuit in Fig (2.6), what effects will this have on the step response?

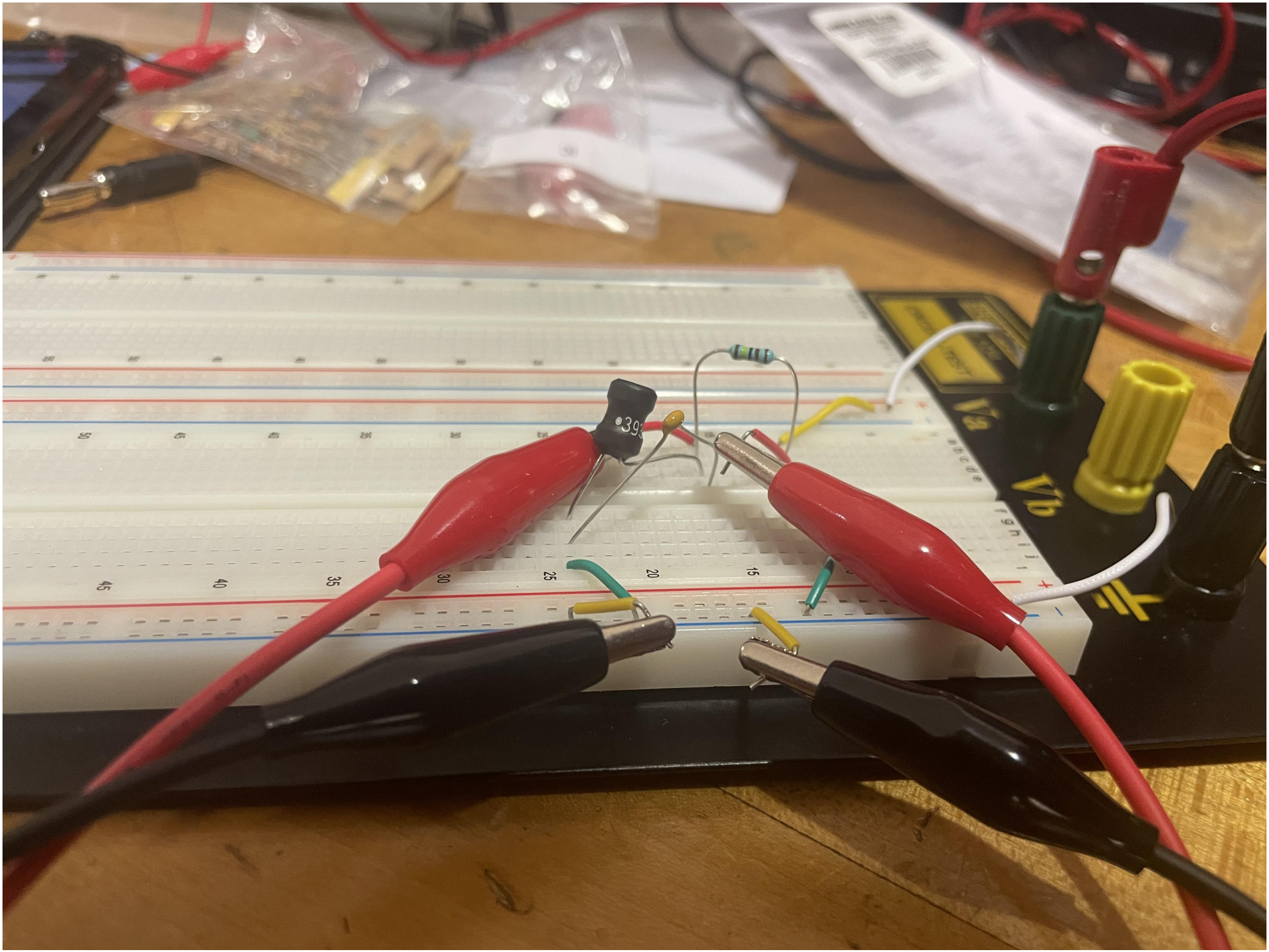
If you remove the resistor, the circuit would become a series L-C circuit with only a 50 Ω resistance providing the dampening. This will cause α to drop from about 6900/s to 625/s. This means the decay would be a lot smaller so the response lasts a lot longer and oscillates a lot more.

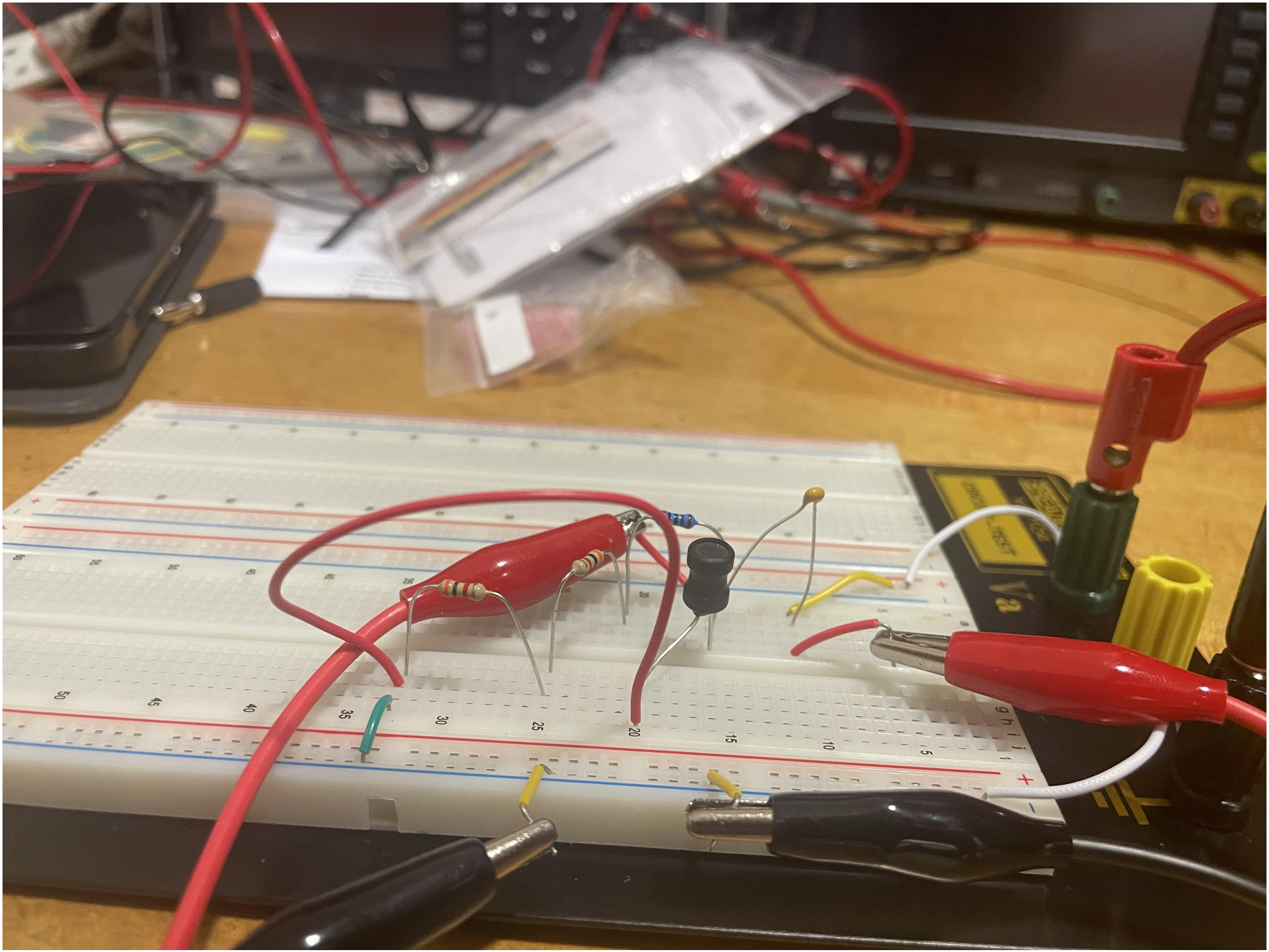

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGA
1 2.00V/ 2 2.00V/
100.0ns
50.00ns/
Auto
1 4.78V
KEYSIGHT TECHNOLOGIES
Cursors
Mode
Track
X1 Source
2
X2 Source
2
Cursors
X2
Units
Back
X1: 0.0s
X2: 127.00000us
ΔX:
+127.000000us
Y1(2):
-165.57mV
Y2(2):
+7.61mV
ΔY:
+173.25mV


BREADBOARD
MB-104
CIRCUIT-TEST
Va
Vb

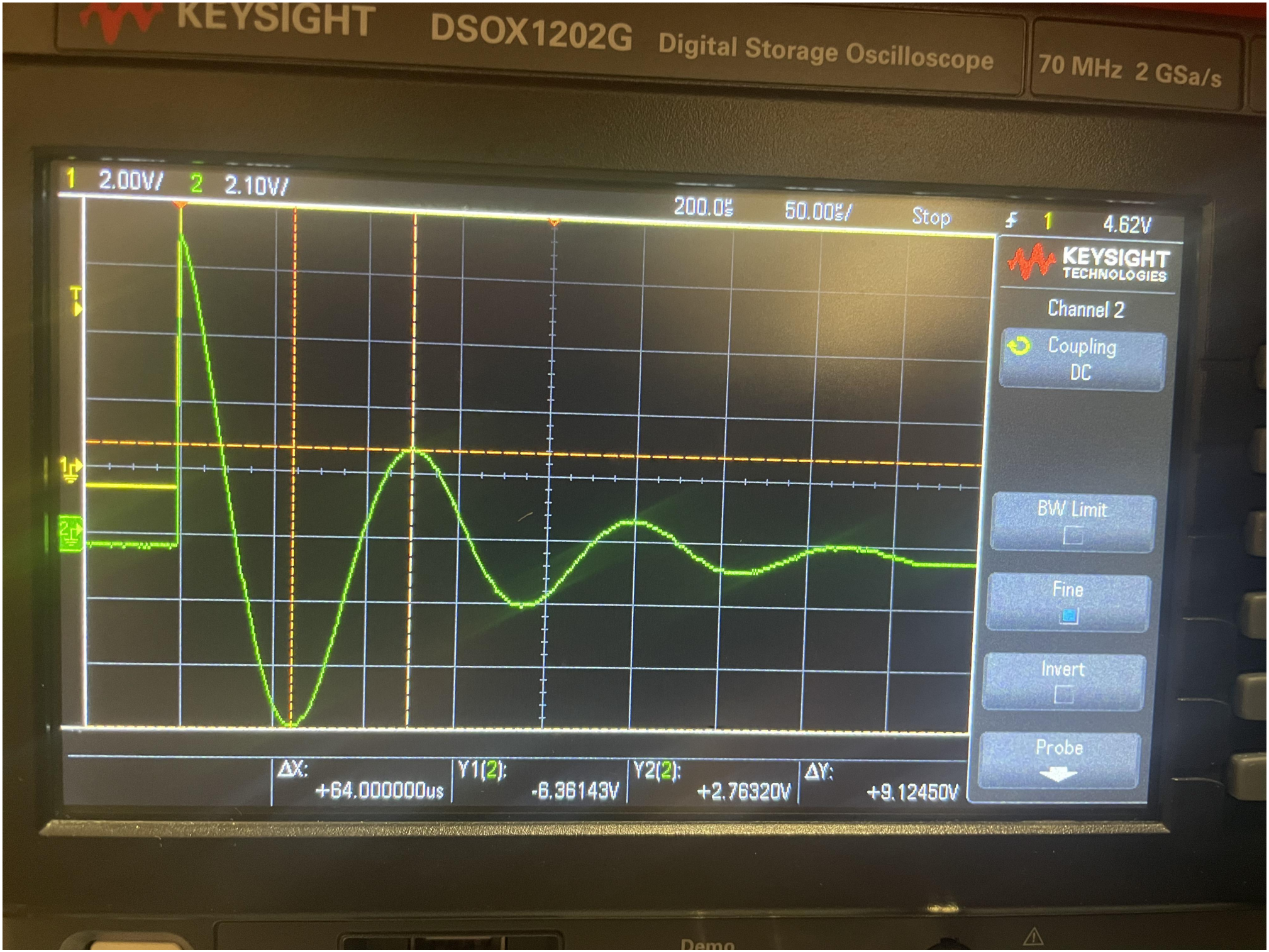

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.00V/
199.0ns
50.00ns/
Auto
1 5.33V
KEYSIGHT TECHNOLOGIES
Cursors
Mode
Track
X1 Source
2
X2 Source
2
Cursors
X2
Units
X1: 0.0s
X2: 136.00000us
ΔX:
+136.000000us
Y1(2):
-163.40mV
Y2(2):
+417.21mV
ΔY:
+580.50mV
Demo
Probe Comp
Gen Out
1


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.10V/
198.0ns
50.00ns/
Auto
1 4.62V
KEYSIGHT TECHNOLOGIES
Channel 2
Coupling
DC
BW Limit
Fine
Invert
Probe
ΔX:
+65.000000us
Y1(2):
-6.04664V
Y2(2):
+3.05311V
ΔY:
+9.09975V


Va
Vb



Vb
Va
MB-104


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.10V/
200.0µs 50.00µs/ Stop
1 4.62V
KEYSIGHT TECHNOLOGIES
Channel 2
Coupling
DC
BW Limit
Fine
Invert
Probe
ΔX: +64.000000us
Y1(2): -6.36143V
Y2(2): +2.76320V
ΔY: +9.12450V